

Activity:

Any process that cause change in the system.

SYSTEM	ENTITIES	ATTRIBUTES	ACTIVITIES
SuperMarket	Staff, Customers	Shopping-list manage items	Buying Selling
Traffic	Vehicles (Bus, Car etc)	Speed, distance covered	Driving Parking
Bank			

System.

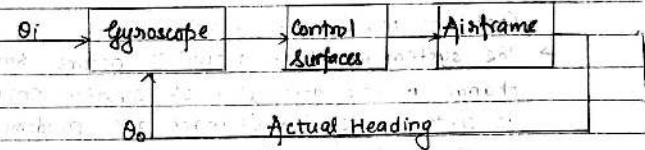
Example

- # Consider an artifact flying through under the control of an autopilot. A gyroscope is the autopilot that detects the difference between actual heading and desired heading. It sends the signal to move the control systems. In response to the control surface movement, the airframe, steers towards the desired heading.

Fig: An aircraft under autopilot control

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Desired Heading.



- # Consider a factory that makes an assembled parts into product. The major component of the system are the fabrication department making the parts and the assembly department maintains a supply of raw materials and a shipping department dispatches the finished products. A production control department receives order and assign work to the other departments.

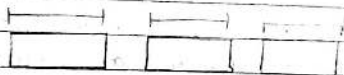
(Table)

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Continuous and Discrete System

→ The system in which activities occurs smooth change in the attributes of system entities is continuous system. Changes are predominantly smooth. E.g:- Turning of aircraft.

→ The system in which changes are predominantly discontinuous are discrete system. E.g:- Banking system



Time interval
(discrete)



(continuous)

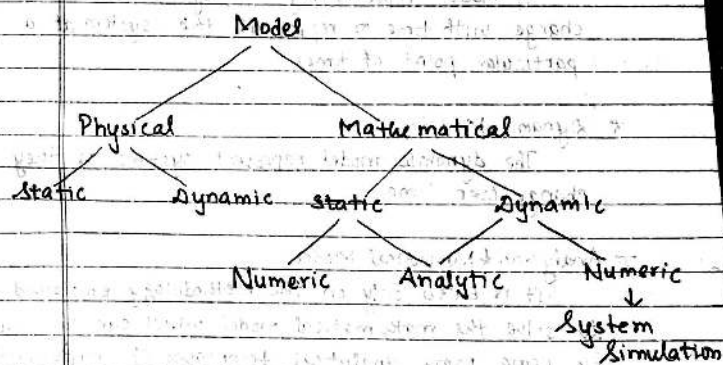
Endogeneous Activities:-

Activities occurring within the system (system [closed])

Exogeneous Activities:-

Activities occurring outside the system which effects the system. (open system)

* Types of Model:



Physical Model.

In this model, physical objects are substituted for real things. This models may comprise of only physical objects without any dynamic interaction between them.

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Mathematical Model:

This model comprise of symbolic notation and mathematical equation to represent the system.

Static Model:

A model representing the system, which doesn't change with time or represent the system at a particular point of time.

Dynamic Model:

The dynamic model represent systems as they change over time.

Analytical & Numerical Model:

It is based only on the methodology employed to solve the mathematical model which can be solved by using analytical techniques of mathematics is analytical model.

The model which requires the application of numerical method are called numerical model.

System Simulation:

Simulation describes as a procedure of establishing a model and deriving solution from it, which covers the whole analysis of physical, analytical &

numerical investigations.

numerical investigations.

Deterministic models:

It has a known set of inputs which results into an unique set of outputs.

Stochastic models:

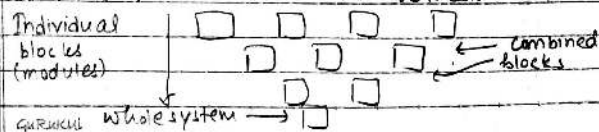
These are one or more random inputs variables, which gives many random outputs. The outputs in such cases are only estimates of true characteristics of system.

* Principles Used in Modeling:

It is not possible to provide exact rules by which models are built but a number of guiding principles can be defined.

a) Block building:-

The description of system should be organized in a series of blocks. Each block describes the part of the system. The system as a whole can then be described in terms of the interconnection between the blocks.



b) **Relevance:** The model should include only those aspects of the system that are relevant or essential to the study objectives. Even though the irrelevant things does not harm the model it increases the complexity.

c) **Accuracy:-**

The accuracy of the information gathered for the model should be considered.

d) **Aggregation:**

Here, the number of individual entities can be grouped together into large entities.

Advantage of simulation:-

- 1) New policies, operating procedures, decision rules, information flow, and so on can be explored without disturbing ongoing operations in the real world.
- 2) Hypothesis about how or why certain phenomena occurred can be tested for feasibility.
- 3) Bottle neck analysis can be performed indicating where in process information.
- 4) "What-If" questions can be answered.

Areas of application:

① **Manufacturing Applications:**

- Analysis of assembly operations
- Analysis of storage and revival strategies in a warehouse
- Investigation of dynamics in a service-oriented supply chain

② **Construction engineering**

- Construction of a dam embankment.
- Activities scheduling in a dynamic, multiproject settings
- Investigation of the structural steel erection process.
- Special purpose template for utility tunnel construction.

③ **Military Application:-**

- Modeling leadership effects and recruits type in an army recruiting station.
- Using adaptive agents in Airforce pilot training.

④ **IT field Applications:**

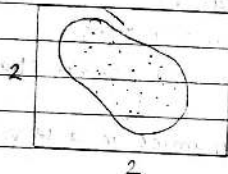
- Testing new designed hardware and software.
- Testing of new generation networking Algorithm.
- Testing of new design mobiles & its applications

Simulation is used in every field now-a-days.

Unit-2

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Monte-Carlo Method:



$$\frac{A}{4} = \frac{8}{20}$$

A = Area of Irregular shape

8 → x = total samples within irregular shape

4 = Area of regular shape

20 = Total no. of samples taken

Calculate the value of π using Monte-Carlo.

Area of circle (A) = πr^2

We know: $x^2 + y^2 = r^2$

$$x^2 + y^2 \leq 1$$

$$y \leq \sqrt{1-x^2}$$

Taking sample points = 20 (N)

$$A = \frac{\pi}{4} = \frac{n}{N}$$

$$A = \frac{1}{4} \pi r^2$$

$$\pi = \frac{4A}{r^2}$$

$$\therefore \pi = 4A$$

$$\frac{\text{Area of circle}}{\text{Area of sq.}} = \frac{\text{samples in shaded area}}{\text{Total no. of samples}}$$

$$\pi = \frac{4n}{N}$$

S.N	x	y	x ²	y ²	Is $y \leq \sqrt{1-x^2}$	Yes/No
1	0.2	0.167	0.04	0.027		Yes
2	0.2	0.32	0.04	0.108		Yes
3	0.2	0.5	0.04	0.25		Yes
4	0.2	0.67	0.04	0.448		Yes
5	0.2	0.83	0.04	0.689		Yes
6	0.4	0.167	0.16	0.027		Yes
7	0.4	0.33	0.16	0.108		Yes
8	0.4	0.5	0.16	0.25		Yes
9	0.4	0.67	0.16	0.448		Yes
10	0.4	0.83	0.16	0.689		Yes
11	0.6	0.167	0.36	0.027		Yes
12	0.6	0.33	0.36	0.108		Yes
13	0.6	0.5	0.36	0.25		Yes
14	0.6	0.67	0.36	0.448		Yes
15	0.6	0.83	0.36	0.689		No
16	0.8	0.167	0.64	0.027		Yes
17	0.8	0.33	0.64	0.108		Yes
18	0.8	0.5	0.64	0.25		Yes
19	0.8	0.67	0.64	0.448		No
20	0.8	0.83	0.64	0.689		No

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$$\pi = \frac{4 \times 17}{20} = 3.4$$

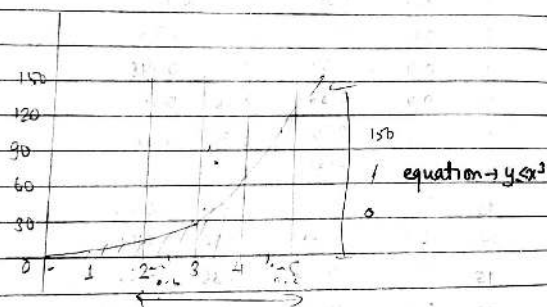
Numerical Integration by Monte-Carlo Method.

$$I = \int_2^5 x^3 dx \text{ using 20 sample points}$$

$$\text{Given } f(x) = x^3$$

$$\text{say } y = x^3$$

x	0	1	2	3	4	5	6
y	0	1	8	27	64	125	216



$$I = \frac{n}{N} \cdot A \rightarrow \text{area of rectangle}$$

area of shaded
ratio of points

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S.No	x	y	x ³	I ₃ (y ≤ x ³)
1	2.5	30	15.625	No
2	2.5	60	15.625	No
3	2.5	90	15.625	No
4	2.5	120	15.625	No
5	3	30	27	No
6	3	60	27	No
7	3	90	27	No
8	3	120	27	No
9	3.5	30	42.875	Yes
10	3.5	60	42.875	No
11	3.5	90	42.875	No
12	3.5	120	42.875	No
13	4	30	64	Yes
14	4	60	64	Yes
15	4	90	64	No
16	4	120	64	No
17	4.5	30	91.125	Yes
18	4.5	60	91.125	Yes
19	4.5	90	91.125	Yes
20	4.5	120	91.125	No

$$n=6, N=20, A=3 \times 150$$

$$I = \frac{6}{20} \times 3 \times 150$$

$$= 135$$

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$$[\ddot{x} = \frac{d^2x}{dt^2}, \dot{x} = \frac{dx}{dt}]$$

Note For better result adjust to make $n=7$ by replace s.no as (4.5, 70). The ans will be near to exact value.

Qr $I = \int_0^4 \sqrt{x} \, dx$

accurate ans	$n=14$ $N=20$ $A=8$	real value = 5.333
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3rd April Comparison of Simulation and Analytical Methods:-

- Simulation gives specific solution than general solution.
- An analytical solution gives all the conditions.
- For e.g.: of Automobile wheel motion ($M\ddot{x} + D\dot{x} + Kx = KFc$)
Analytical gives all condition that can cause oscillation but simulation tells only whether a particular set of condition did or did not cause oscillation.
- To find all such condition simulation must be repeated under different conditions.
- To judge which method to apply, it is necessary to consider carefully what question need to be answered in the system study and to what degree of accuracy the answer need to be known.
- The amount of detail in the model need to be limited to minimum level in simulation.
- The ideal way of using simulation is an extension of mathematical solutions.

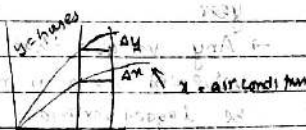
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- There are certain limits in Analytical method like physical stops, finite time delay, non-linear forces etc.
- Such limitations are removed by simulation.
- Simulation method provides a quicker or more convenient way of deriving results than analytical methods.
- It is more convenient to use simulation to obtain results directly.

Experimental Nature of Simulation.

- Simulation is essentially an experimental problem-solving techniques.
- Many simulation runs have to be made to understand the relationship involved in the system.
- So, the use of simulation in a study must be planned as a series of experiment.
- To find the output or result in simulation the experiment or test or procedure must be implemented or run to we can see the appropriate output.

Types of System Simulation:



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Distributed Lag Model:

Models that have the properties of changing only at fixed intervals of time and of basing current values of the variable on other current values and values that are occurred in previous intervals are called distributed lag models.

They are used extensively in econometric studies.

Let C = consumption

I = Investment

T = Taxes

G = government expenditures

Y = National income.

$$C = 20 + 0.7(Y - T)$$

$$I = 2 + 0.1Y$$

$$T = 0.2Y$$

$$Y = C + I + G$$

This is a static mathematical model, but it can be made dynamic by taking a fixed interval, say one year.

Any variable that appears in the form of its current value and one or more previous intervals is said to be lagged variable.

The set of equation can be written in the form

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$$I = 2 + 0.1Y_{-1}$$

$$Y = 45.45 + 2.27(I + G)$$

$$T = 0.2Y$$

$$C = 20 + 0.7(Y - T)$$

Q. Year	G
1	20
2	25
3	30
4	35
5	40

$$Y_{-1} = 80$$

For 1st Year:

$$I = 2 + 0.1Y_{-1}$$

$$= 2 + 0.1 \times 80$$

$$= 10$$

$$\begin{aligned} Y &= 45.45 + 2.27(I + G) \\ &= 45.45 + 2.27(10 + 20) \\ &= 113.55 \end{aligned}$$

$$T = 0.2Y$$

$$= 0.2 \times 113.55$$

$$= 22.71$$

$$C = 20 + 0.7(Y - T)$$

$$= 20 + 0.7(113.55 - 22.71)$$

$$= 88.588$$

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For 2nd Year.

$$Y_{-1} = 113.55$$

$$\begin{aligned} I &= 2 + 0.1 \times Y_{-1} \\ &= 2 + 0.1 \times 113.55 \\ &= 13.355 \end{aligned}$$

$$\begin{aligned} Y &= 45.45 + 2.23(I + T) \\ &= 45.45 + 2.23(13.355 + 25) \\ &= 132.51585 \end{aligned}$$

$$\begin{aligned} T &= 0.2 \times Y \\ &= 0.2 \times 132.51585 \\ &= 26.50317 \\ &\approx 26.5 \end{aligned}$$

$$\begin{aligned} C &= 20 + 0.7(Y - T) \\ &= 20 + 0.7(132.51585 - 26.50317) \\ &= 94.2 \end{aligned}$$

For 3rd Year

$$Y_{-1} = 132.515$$

$$\begin{aligned} I &= 2 + 0.1 \times 132.515 = 15.25 & T &= 29.632 \\ Y &= 148.16 & C &= 102.9696 \end{aligned}$$

Guaranteed

For 4th Year

$$Y_{-1} = 148.16$$

$$\begin{aligned} I &= 2 + 0.1 \times 148.16 = 16.816 \\ Y &= 45.45 + 2.23(16.816 + 35) = 163.07 \\ T &= 0.2 \times 163.07 = 32.61 \\ C &= 20 + 0.7(163.07 - 32.61) = 111.822 \end{aligned}$$

For 5th Year

$$Y_{-1} = 163.07$$

$$\begin{aligned} I &= 2 + 0.1 \times 163.07 = 18.307 \\ Y &= 45.45 + 2.23(18.307 + 40) = 177.80 \\ T &= 0.2 \times Y = 35.56 \\ C &= 20 + 0.7(177.80 - 35.56) = 119.56 \end{aligned}$$

5th April

Cobweb model:

Distributed lag model can be constructed from:

- The static model.
- This relates supply (S), demand (D) and market price (P).
- The supply should be made dependent upon the price from the previous marketing period.
- The demand will respond to current price.
- Again assuming the market is cleared, the market in distributed lag form is:

Guaranteed

So Supply, $D = \text{Demand}$, $P = \text{Market Price}$.

$$Q = a - bP \quad \begin{matrix} Q = \text{Quantity} \\ S = c + dP \quad P = \text{Price} \\ Q = S \end{matrix}$$

Initial value of P_0 will be given

① Fluctuation of Market Price

Given data:-

$$P_0 = 1.0$$

$$a = 12.4$$

$$b = 1.2$$

$$c = 1.0$$

$$d = 0.9$$

When $T=0$, $P_0 = 1.0$

$$Q = 12.4 - (1.2 \times 1.0) = 12.4 - 1.2$$

$$Q = 11.2$$

$$S = 1.0 + (0.9 \times 1.0) = 1 + 0.9 = 1.9$$

$$S = 1.9$$

$$Q = S = 1.9$$

Now, $1.9 = 12.4 - (1.2 \times P_1)$ $P_1 = ?$

$$P_1 = 8.75 \leftarrow \frac{12.4 - 1.9}{1.2}$$

$$S = 1.0 + (0.9 \times 8.75) = 1 + 7.875 = 8.875$$

$$S = 8.875$$

$$Q = S = 8.875$$

Also,

$$8.875 = 12.4 - (1.2 \times P_2)$$

$$P_2 = \frac{12.4 - 8.875}{1.2} = 2.9375$$

Given data

$$P_0 = 5$$

$$a = 10$$

$$b = 0.9$$

$$c = 2.4$$

$$d = 1.2$$

$$S = 1.0 + 0.9 \times 2.9375 = 3.64375$$

$$Q = S = 3.64375$$

$$\text{So, } 3.64375 = 12.4 - (1.2 \times P_3)$$

$$P_3 = 7.2968$$

$$S = 1.0 + 0.9 \times 7.2968$$

$$S = 7.567188$$

$$\text{Again, } 7.567188 = 12.4 - (1.2 \times P_4)$$

$$P_4 = 4.027$$

$$S = 1.0 + 0.9 \times 4.027$$

$$= 4.624609$$

$$Q = S = 4.624609$$

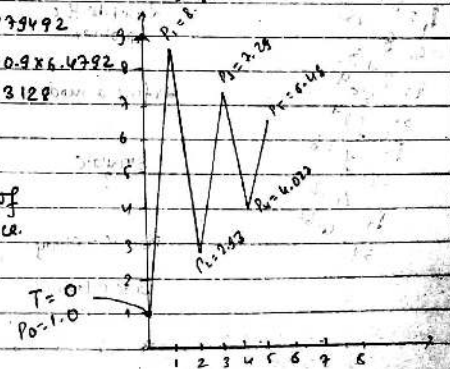
$$\text{Again, } 4.624609 = 12.4 - (1.2 \times P_5)$$

$$P_5 = 6.479492$$

$$S = 1.0 + 0.9 \times 6.479492$$

$$S = 7.83128$$

Fig: Fluctuation of market price.



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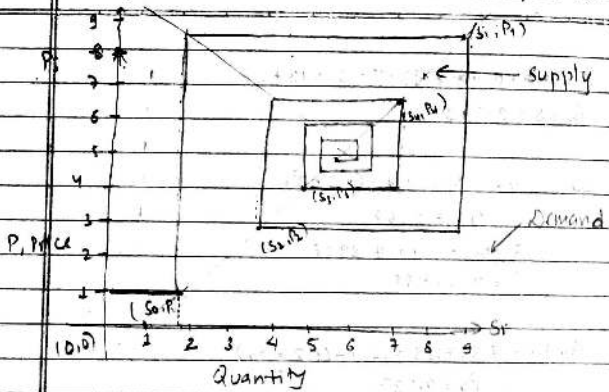
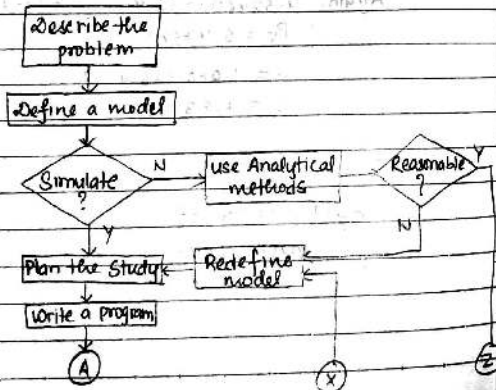


Fig: CobWeb graph.
Relation of demand and supply

Steps in a Simulation Study:-



25th April.

Time Advancement Mechanism:

- is a component of any simulation model.
- while performing simulation after each event is executed, the time goes forward to find the state of the system at a new point of time.
- keeps track of passed time and the action or experiment at the end of the specific simulation period.
- also called time flow mechanism, timing routines or simulation executives.

Two types:

(1) Fixed Time Increment Methods:

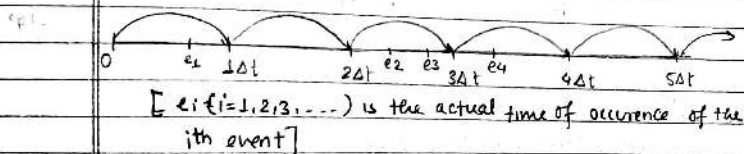
- also called time oriented simulation.
- a timer is simulated which is advanced by a fixed time.

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→ After each increment see if it should have each event type is checked during the time of the last increment.

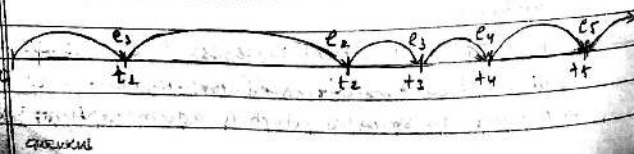
increment.

- after each fixed interval, the system is scanned to identify the event that occurs during incremental time.
- all of the events are treated as if they occurred at the end of the increment time.
- used in both continuous and discrete simulations.



② Next Event Increment Methods:

- also called Event Oriented simulation
- timer is advanced to event to event.
- state of the system is updated at each event.
- used in all discrete-event simulation models.
- here the simulation clock is initialized to zero and the times of occurrence of future events are determined.



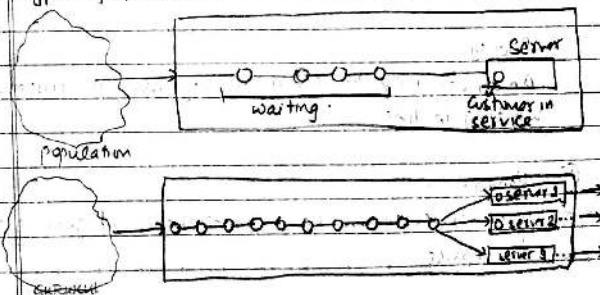
Queuing Model:

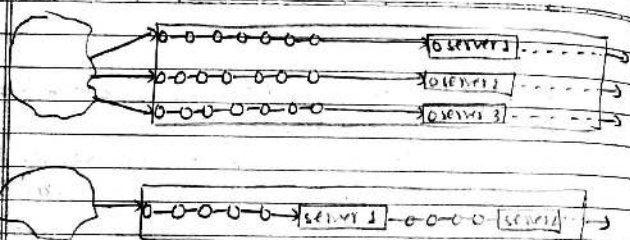
- Waiting lines or queues are a common site in Real Life.
- This situation occurs because:
 - ① There is too much demand on the service facility. So, customer or entities or events have to wait for getting services.
 - ② There is too less demand, where service facility have to wait for the entities or customers to come.

Components:

- ① Calling sources: Population from where customer are drawn or called.
- ② Waiting Line or queue: The number of customers waiting to be served.
- ③ Service facility: The number of service (counter) channels.

Types of Queues:





29th July:

Queue Discipline:

→ It determines how the customers are selected from the queue for services

① First-In-First Out (FIFO):

→ Also called as First ^{come} First served

→ Customers are served in the order of their arrivals

② Last-In First Out (LIFO):

Here the last arriver is first served.

③ Priority:

→ An arrival may be given priority over the customer waiting in line

④ Random:

→ When all customers have equal chance of getting selected for service.

Continued

① FIFO

e.g.:- Queue in ticket counter.

② LIFO

e.g.:- Stack.

③ Priority

e.g.:- Hospitals.

④ Random

e.g.:- Food Counter.

Measures of queue:-

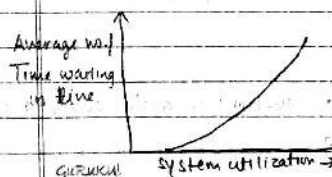
The performance of a queuing system can be evaluated in terms of a number of response parameters as follows:

1) Average no. of customers in the queue or in the system.

2) Average waiting time of a customer in the queue or in the system.

3) System utilization

→ The cost of waiting time & idle time.



System utilization factor(s) = $\frac{\lambda}{\mu}$ in single server

(s) = $\frac{\lambda}{m\mu}$ in multiple server.

λ = average arrival rate
 μ = average service rate.

Continuous System:

- A continuous system is one in which the predominant activities of the system cause smooth changes in the attributes of the system entities.
- When such system is modeled mathematically, the variables of the models representing the attributes are controlled by continuous functions.
- Mathematical model consists of differential equations.

Q. * Differential Equations:

E.g:- $M\ddot{x} + D\dot{x} + Kx = K F(t)$

M, D, K = constants

x = dependent variable

t = independent variable

Analog Computer:

To represent differential equations in analog computer many different symbols are used.

Continued

ject:

Assignment

RL circuit
 RC "

$E(t) = R\dot{q} + L\ddot{q}$
 $E(t) = R\dot{q} + \frac{1}{C}q$

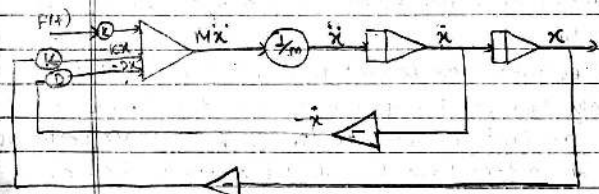
summer (∇) = for adding, $\triangleright$ sign changer.

$\bigcirc$ → scale factor

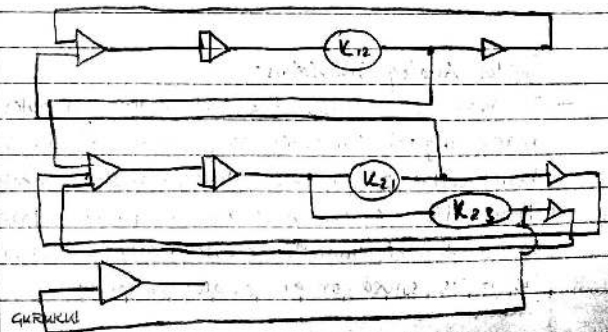
$\square$ Integrator

$M\ddot{x} = (KF(t) - D\dot{x} - Kx)$

$\ddot{x} = \frac{1}{m} (\quad)$



Ind May: Draw Analog Computer



$$\dot{x}_1 = -k_{12}x_1 + k_{21}x_2$$

$$\dot{x}_2 = k_{12}x_1 + (k_{21} + k_{23})x_2$$

$$\dot{x}_3 = k_{23}x_2$$

Hybrid Computers:-

- Hybrid computers describe the combination of transition of analog-computer elements giving smooth, continuous outputs and elements carrying out such non-linear digital operations as storing values, switching & performing logical operations.
- This computer may be used to simulate systems that are mainly continuous do-infact have some digital elements. Eg:- an artificial satellite, for which both the continuous but do in fact have some digital elements. Eg:- an artificial satellite for which both the continuous eqⁿ of motion and the digital control system.

Digital Analog Simulators: (analog → analog → digital)

- To avoid the disadvantages of analog computer, many digital computer programming languages have been written to produce digital-analog simulators.
- They allow a continuous model to be programmed on a digital computer in essentially the same way as it is solved on an analog computer.

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- The languages contain micro-instruction that carry out the action of adder, integrators & sign changers.

CSSL (Continuous System Simulation Language)

- They use the familiar statement-type of input for digital computer, allowing a program problem to be programmed directly from the equation of a mathematical model.
- CSSL include a variety of algebraic & logical expressions to describe the relations between variables.

CSMP-III (Continuous System Modeling Program Version-3)

- A CSMP-III program is constructed from three general type of statements:

① Structural statements:-

It defines the model. They consist of FORTRAN like statements and functional blocks designed for operations that frequently occur in the model definitions.

② Data statements:-

- which assigns numerical values to parameters, constants, and initial conditions.

③ Control statements:-

- which specify options in the assembly and execution of the program, and the choice of output.

Structural statement can make use of the operation of

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addition, subtraction, multiplication, division and exponential, using the same notation and rules as used in FORTRAN.

E.g:- $X = 6Y + (Z-2)^2$

$= X = 6.0 * Y / W + (Z-2.0) ** 2$

$Y = \text{INTGRL}(IC, X) \quad Y = \int_0^1 x \, dt + IC$

$Y = \text{EXP}(X) \quad Y = e^x$

$Y = \text{SIN}(X) \quad Y = \sin(x)$

$Y = \text{ALOG}(X) \quad Y = \ln(x)$

$Y = \text{ABS}(X) \quad Y = |x|$

e.g:- $M\ddot{x} + D\dot{x} + kx = K F(t)$

CSMP-III Program.

TITLE Automobile Suspension System.

*

PARAM: D = (5656, 16.968, 39.592, 56.56, 113.12)

X2DOT = (1.0/M) * (K * F - K * X - D * XDOT)

XDOT = INTGRL(0.0, X2DOT)

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in $X = \text{INTGRL}(0.0, XDOT)$

*

CONST M=2.0, F=1.0, K=400.0

TIMER DELT=0.005, FINITIM=1.5, PRDEL=0.05, OUTDEL=0.05

PRINTX, XDOT, X2DOT

PRTPLT X

LABEL DISPLACEMENT VERSUS TIME

END

STOP

DELT: Integration Interval

FINITIM: Finish Time

PRDEL: Interval at which to print result

OUTDEL: Interval at which to print plot

Hybrid Simulation:

Systems are either continuous or discrete in nature.

But, there are times when an analog and digital computer are combined to provide a hybrid simulation.

It is also possible that the system being simulated is an interconnection of continuous & discrete sub-systems, which can be modeled by an analog and digital computer being linked together.

Here, high-speed converters are needed to transform signals from one form to another.

GURUKUL

Page 75-76

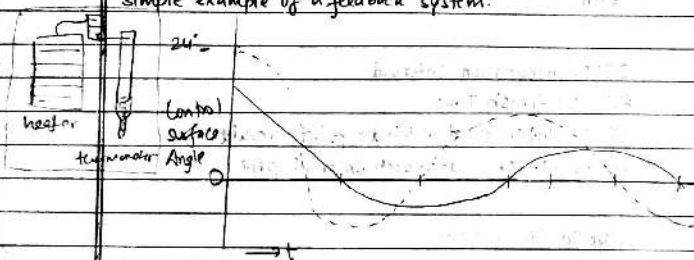
Date: _____ Page: _____

Eg:- Satellite System
Mobile Communication.

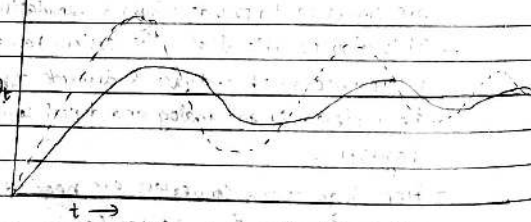
Feedback System

In this system, coupling occurs between the input and output of the system. The term feedback is used to describe the phenomenon.

A home heating system controlled by a thermostat is a simple example of a feedback system.



Aircraft
Heading



CHURCHILL

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Interactive System

- is concerned with the response of a system over time.
- the observation of the response over time is important for continuous system simulation.
- A user can judge whether the output is developing satisfactory and if not interrupt a simulation run to try for better conditions.

Real Time Simulation

- Here preliminary work could be avoided if an actual device can be used rather than constructing a model.
- This approach followed when using real time simulation.
- With this technique, actual devices, which are part of system, are used in conjunction with either a digital or hybrid computer, providing a simulation of the part of the system that do not exist or that cannot conveniently be used in experiment.
- The computer model should run at the same rate as the actual physical system runs.
E.g:- In computer gaming.
- The aerospace industry in particular makes extensive use of real time simulation.
- There are devices called simulators, whose main function is to provide human beings with a substitute for some environment or situation.

CHURCHILL

Training of pilots before using actual plane.
~~Training for~~

Predator-Prey Model

Let $x(t)$ = Number of individual in the prey at the time t
 $y(t)$ = Number of individual in the predator at time t

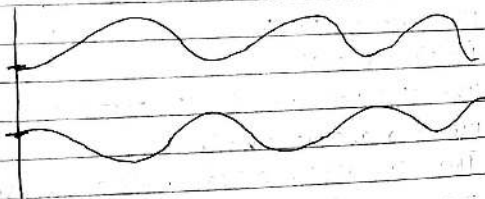
Suppose now there is sufficient food (grass) for prey animals & in the absence of predator, here,
 $r x(t)$ = rate of growth of prey animals.

(We take 'r' as: natural birth rate - natural death rate)

The death rate of the prey due to interaction with predator can be assumed to be proportional to the product of two population sizes, $x(t) \cdot y(t)$

$\therefore$ The overall rate of change of the prey population is given by:

$$\frac{dx}{dt} = r x(t) - a \cdot x(t) \cdot y(t), \quad a = \text{positive constant of proportionality.}$$



Subject:

Chapter:

Date: 13th May Page:

Random Numbers:

Random Numbers are a necessary basic ingredients in the simulation of almost all discrete system. Most computer languages have a subroutine, object or function that will generate a random numbers.

Similarly, simulation language generate random numbers that are used to generate event times & other random variables.

Properties:-

A sequence of random numbers has two important properties.

- (1) Uniformity
- (2) Independence.

Each random number is an independent sample drawn from a continuous uniform distribution between an interval 0 to 1. The probability density function is given by:

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

The expected value of each Random Number R_i is given by

$$E(R) = \int_0^1 x \cdot dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}$$

Quaruch

and its variance is given by:

$$V(R) = \int_0^1 x^2 dx - E(R)^2 = \frac{x^3}{3} \Big|_0^1 - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{3} - \frac{1}{4}$$

$$V(R) = \frac{1}{12}$$

Random Number Table.

Table gives random number produced by certain physical process. The digits have been arranged in column and rows.

Pseudo Random Numbers:

The word Pseudo means false. But here word pseudo implies that the random number are generated by using some kind of known arithmetic ^{operations} process. If the method is known, the set of random number are not truly random. So we call pseudo random numbers. However, the pseudo random numbers generated by many computer routines vary closely fulfill the requirement of desired randomness.

Some problems while generating pseudo random numbers:

- ① The generated ~~num~~ numbers may not be uniformly distributed
- ② The generated number may not be continuous.

③ The mean and variance of generated numbers may be too high or low.

④ There may be cyclic pattern in the generated number.

Generation of Random Numbers:

Congruence Method or Residue Method:
Expressed by: $r_{i+1} = (ar_i + b) \text{ MOD } m$

where a, b, m are constants.

r_{i+1} & r_i are the $(i+1)$ and i th term random numbers.

③ Mixed Multiplicative Congruential (MMCG) method
 $r_{i+1} = (ar_i + b) \text{ MOD } m$

Given:

$$a = 13$$

$$b = 1$$

$$m = 19$$

$$r_0 = 1$$

$$r_1 = (a r_0 + b) \text{ mod } m$$

$$= (13 \times 1 + 1) \text{ mod } 19$$

$$= 14$$

$$r_2 = (13 \times 14 + 1) \text{ mod } 19$$

$$= (182 + 1) \text{ mod } 19$$

$$= 17$$

$$r_3 = (13 \times 17 + 1) \text{ mod } 19$$

$$= (221 + 1) \text{ mod } 19$$

$$= 18$$

$$r_3 = 3$$

$$r_4 = 2$$

$$r_5 = 6$$

$$r_6 = 1$$

$$r_7 = 12$$

b) Multiplicative congruential (MC) method

$$r_{i+1} = ar_i \text{ mod } m$$

find same as $r_1, r_2, r_3, \dots$

c) Additive

$$r_{i+1} = (r_i + b) \text{ mod } m$$

Testing of Randomness:

Uniformity Testing

① The Kolmogorov-Smirnov Test

② Chi-squared test

① The Kolmogorov-Smirnov Test

Steps:

1) Rank the data from the smallest to largest

Let $R(i)$ denote i th smallest observation

Given

$$R(1) \leq R(2) \leq R(3) \leq \dots \leq R(N)$$

3) Calculate:

$$D = \max \{ D^+, D^- \}$$

3) Calculate

$$D^+ = \max \{ i/N - R(i) \}$$

$$D^- = \max \{ R(i) - \frac{i-1}{N} \}$$

4) Determine the critical value from standard table chart.

5) If the sample statistic D is greater than the critical value D_α then it is rejected. If D is smaller than D_α then it is accepted.

Q) Given Random numbers 0.44, 0.83, 0.14, 0.05, 0.93. Perform a K-S test with a level of significance α is 0.05.

Given, $N=5$

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
$R(i)$	0.05	0.14	0.44	0.83	0.93
i/N	$1/5$	$2/5$	$3/5$	$4/5$	$5/5$
$D^+ = i/N - R(i)$	0.15	0.26	0.16	-0.01	0.07
$D^- = R(i) - \frac{i-1}{N}$	0.05	-0.06	0.04	0.21	0.13

$$D^+ = 0.26$$

$$D^- = 0.21$$

So,

$$D = 0.26$$

critical value $D_{0.05} = 0.565$ (from table).

$D < D_{0.05}$ So accepted.

QURRACH

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Chi-squared Test: (Uniformity Testing)

→ The chi-square test uses the sample statistics:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

O_i = Observed number in the i th class

E_i = Expected number in the i th class

n = number of class.

Degree of Freedom (DoF) = $n - 1$

Q. Use the chi-square test with 2 of 95% to test the data shown below are uniformly distributed.

10	63	5	64	18	6	9	1	17	21
31	40	99	20	30	55	2	27	42	58
81	8	7	9	21	22	22	82	97	9
4	16	28	38	41	52	68	71	86	92
3	33	24	7	53	59	25	13	46	16
25	2	34	58	8	24	50	26	14	92
58	23	3	23	35	9	43	12	55	15
81	22	28	4	36	18	10	37	30	72
21	83	27	43	5	19	32	11	87	95
12	41	84	29	70	6	53	31	12	56

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Interval	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
1 - (1-10)	21	10	11	121	$\frac{121}{10} = 12.1$
2 - (11-20)	15	10	5	25	2.5
3 (21-30)	15	10	5	25	2.5
4 (31-40)	19	10	9	81	8.1
5 (41-50)	6	10	-4	16	1.6
6 (51-60)	10	10	0	0	0
7 (61-70)	3	10	-7	49	4.9
8 (71-80)	6	10	-4	16	1.6
9 (81-90)	7	10	-3	9	0.9
10 (91-100)	7	10	-3	9	0.9
	180		0		27.5

$$100 - 95 = 5\%$$

$$\frac{5}{100} = 0.05$$

$$DoF = 10 - 1 = 9$$

Chi-Square Distribution with degree at 0.05

$$D = 15.16$$

$D < \text{Table value}$

$\therefore H_0$ is not rejected

CHURCHILL

Date: 10th May Page: _____

Consider 10×10 table.

biggen gap

(10, 63) 142 11

(10, 63) 142 11

[illegible]

$\alpha = 0.5$

9-2 = 7

$$\alpha^2 \angle D \rightarrow \text{Ac.}$$

→ Discrete Events: A number used to represent some aspect of the system state is called a state description.

→ We define discrete event as a set of circumstances that cause an instantaneous change in one or more system

→ We define discrete event as a set of circumstances that cause an instantaneous change in one or more system state descriptors.

→ The passage of time is recorded by a number & subsequently indicates how many units of simulation time have passed since the beginning of the simulation.

Two basic method exists for updating clock time.

- ① To advance the clock to the time at which next event due to occur. This is referred as event oriented.
- ② To advance clock by small (uniform) interval of time.

Simulation of a Telephone system.

	1	2	3
1			
2		•	
3			•
4		•	

From	To	Length
1	7	20

Arrival Time
1057

Lines

1	0
2	1
3	0
4	1
5	1
6	0
7	1
8	0

Max No 3
In use 2

Call in progress

Clock

1027

From	To	End
4	7	1045
2	5	1053

Processed	Completed	Blocked	Busy
132	99	5	28

Next call

From	To	Length
1	7	20

Arrival Time
1057

Lines

1	0
2	0
3	0
4	1
5	0
6	0
7	1
8	0

Links

Max no 3
In use 1

Clock

1053

From To End

From	To	End
4	7	1075

Processed	Completed	Blocked	Busy
132	99	5	28

Next call

From	To	Length
1	7	20

Arrival Time
1057

Lines

1	0
2	0
3	0
4	1
5	0
6	0
7	1
8	0

Links

Max no 3
In use 1

Clock

1053

From To End

From	To	End
4	7	1075

Processed	Completed	Blocked	Busy
132	99	5	28

Next call

From	To	Length
1	7	20

Arrival Time
1057

Lines

1	0
2	0
3	0
4	1
5	0
6	0
7	1
8	0

Links

Max no 3
In use 1

Clock

1053

From To End

From	To	End
4	7	1075

Processed	Completed	Blocked	Busy
132	99	5	28

Next call

From	To	Length
1	7	20

Arrival Time
1057

Lines

1	0
2	0
3	0
4	1
5	0
6	0
7	1
8	0

Links

Max no 3
In use 1

Clock

1053

From To End

From	To	End
4	7	1075

Processed	Completed	Blocked	Busy
132	99	5	28

Next call

From	To	Length
3	6	98

Arrival Time
1063

Lines

1	0
2	0
3	0
4	1
5	0
6	0
7	1
8	0

Links

Max no 3
In use 1

Clock

1057

From To End

From	To	End
4	7	1075

Processed	Completed	Blocked	Busy
132	99	5	29

Next call

From	To	Length
1	5	132

Arrival Time
1082

Lines

1	0
2	0
3	1
4	1
5	0
6	1
7	1
8	0

Links

Max no 3
In use 2

Clock

1063

From To End

From	To	End
3	6	1161
4	7	1075

Processed	Completed	Blocked	Busy
132	99	5	29

Next call

From	To	Length
1	5	132

Arrival Time
1082

Lines

1	0
2	0
3	1
4	1
5	0
6	1
7	1
8	0

Links

Max no 3
In use 2

Clock

1063

From To End

From	To	End
3	6	1161
4	7	1075

Processed	Completed	Blocked	Busy
133	99	5	29

A)

Lines	Next Call	From To length	Arrival Time	Delay call	From To length
1	0	3	0	98	
2	0				
3	0				
4	1	Max no	3		
5	0	In use	1		
6	0				
7	1	Clock	1057		
8	0				

Processed	Completed	Blocked	Busy
133	99	5	29

Fig: System state - 3A [From last call]

Delay

B)

Lines	Next Call	From To length	Arrival Time	Delay call	From To length
1	0	3	0	98	
2	0				
3	1	Max No	3		
4	1	In Use	2		
5	0				
6	1	Clock	1063		
7	1				
8	0				

System state
SA

Processed	Completed	Blocked	Busy
133	99	5	29

C)

Lines	Next Call	From To length	Arrival Time	Delay call	From To length
1	1	3	0	98	
2	0				
3	1	Max No	3		
4	0	In use	2		
5	0				
6	1	Clock	1075		
7	1				
8	0				

System state
SA

QUERICK

Processed	Completed	Blocked	Busy
133	100	5	29

22nd May

Gathering Statistics:

Most simulation programming system includes a report generator to print out statistics gathered during the run. The exact statistics required from a model depends upon the study being performed.

The commonly needed statistics are:-

- Counts:** It gives the number of entities of a particular type or the number of times some event occurred.
- Summary measures:** It measures such as extreme values, mean values and standard deviation.
- Utilization:** It is defined as the fraction (or percentage) of time some entity is engaged.
- Occupancy:** It is defined as the fraction (or percentage) of a group of entities in use on the average.
- Distribution:** It is of important variable such as queue lengths or waiting times.
- Transit times:** It is defined as the time taken for an entity to move from one part of the system to another part.

Chapter:

B.D → Blocks represent activities
 This joining the blocks, indicate the sequence.
 in which the ach can be exec.

General Purpose Simulation System (GPSS):-

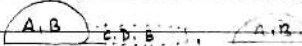
- It is a process oriented language designed for discrete event modeling.
- It was designed by Geoffrey Gordon in 1961 at IBM.
- It has in-built mechanisms to collect statistics, analyze them, produce tabulated outputs & execute a number of different tasks.

GPSS Block Diagram

- The development of a simulation model in GPSS is a block by block construction.
- A set of standard block is arranged in the form of block diagram that represent the flow of entities through the various parts of the system.
- Each block represent a step in a action of the system and links, representing the sequence of events that can occur.
- Each block must be assigned a block time.
- Each block can be used any number of times in a block diagram of a system.
- Total number of blocks used in a system should not be more than 2047.

→ There are about 48 specific block types.

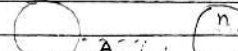
Block Types: 1. Generate, 2. Terminate, 3. Hold, Seize, Release, 4. Queue & Depart, 5. Advanced, 6. Store, Enter, Leave, 7. Mark & Tabulate.

① Generate: 

→ used for creation of transaction

→ employs action time.

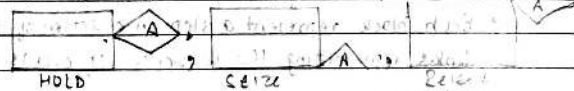
Syntax: GENERATE A, B.

② Terminate: 

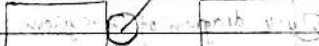
→ is concerned with the removal of the transactions.

Syntax: TERMINATE A.

③ HOLD, SEIZURE, RELEASE



→ are concerned with the use of facilities.

④ QUEUE & DEPART: 

Queue Depart

→ Queue Symbol: is used at the position where over-crowding in flow of transaction is expected.

→ Depart is used for the release of queued object.

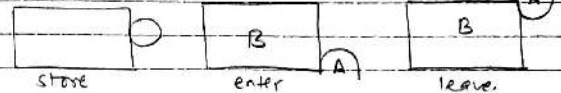
Queue

⑤ ADVANCED: 

→ It is concerned with the flow of transactions.

→ It represents any action required time.

⑥ STORE, ENTER, LEAVE:

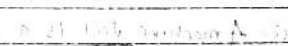


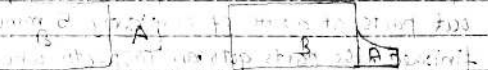
→ The flag on side of each block is for indicating the number of store.

→ STORE block allows a transaction to occupy space in the system.

→ ENTER & LEAVE block allows a transaction to occupy a space or vacate a space in store.

→ ENTER will allow a transaction to enter a store if space is available.

⑦ MARK & TABULATE: 



Mark Tabulate

→ These blocks are used to measure the time spent by a transaction in the system or part of the system.

Queue

⑧ BRANCH: S-1

It is concerned with the movement of transaction & 127 exists are possible

⑨ TRANSFER:



Block 1 Block 2

is used when the transaction are moved in a non-sequential manner

e.g.:- TRANSFER 0.1, ACC, BLOCK 2

⑩ SIMULATE, START, END:

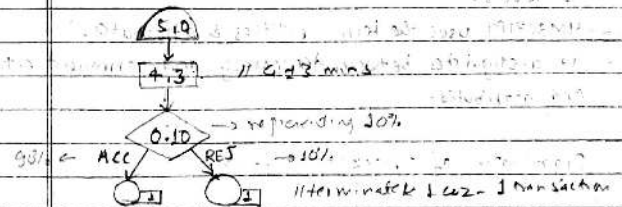
- SIMULATE is generally first statement of the program
- START will start the simulation
- END is always the last statement

Q> A machine tool is a manufacturing shop is turning out parts at a rate of one every 5 minutes. As they are finished, the parts go to an Inspector who takes 4 ± 3 minutes to examine each parts and rejects about 10% of the parts. Each part will be represented by one transaction and the time unit selected for the problem will be 1 minute. Draw GPSS block diagram & write

File new model:
Student G/S eq.

edit & connect gps

GPSS program.



SIMULATE

GENERATE 5.0

ADVANCE 4.3

TRANSFER 0.1, ACC, REJ

ACC TERMINATE 1

REJ TERMINATE 1

START 100

END

26th May SIMSCRIPT

SIMSCRIPT is a very widely used language for simulating discrete systems.

- There are different versions of SIMSCRIPT
- The language be considered as more than just a simulation language since it can be applied to a general programming problems.
- The description of the language given is organized in

5 levels.

- SIMSCRIPT uses the term entities & attributes.
- It distinguishes between temporary and permanent entities and attributes.

Organization of SIMSCRIPT Program:

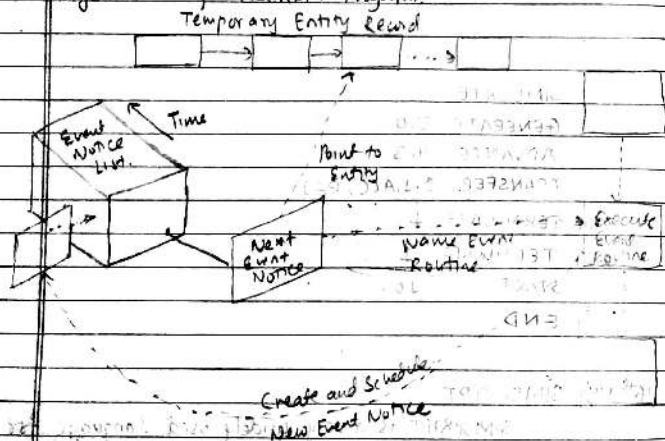


Fig: SIMSCRIPT Execution Cycle.

Subject:

Name for documentation purposes name can be of any length whereas for specific machine implementation the name should consist of an no. of initial char. (typically 8-10).

Names and Labels:

- The user must describe all the entities by giving name to each entity & its attributes.
- Name may consist of any combination of letters & digits provided there is at least one letter.
- We can use dot operator also.
- E.g:- PERSON
- PERSON-AGE
- Labels used for identifying programming statement similarly consists of any combination of letters & numbers without the restriction that at least one be a letter.

Event Routine SIMSCRIPT statements are written in a form closely resembling the English language.

→ A statement calling 'n' lines of the text to be printed could be written in either of the ways.

PRINT n LINES AS FOLLOWS:

PRINT n LINES THUS

PRINT n LINES WITH X AND Y LIKE THIS

X&Y represent the variables or expression to be printed.

E.g: A 3 digit integer indicated by ***

" 4 " real number " ***.***

Date: 30th May Page: _____

Defining the System:

A SIMSCRIPT program begins with a preamble, that defines the system structure and establishes the condition under which the simulation will be run. Variables can be real, integer or alphabetical.

- It is possible to pack variables so that more than one value is placed in one word.

- It is possible to process character strings, called *look words of operation*.

Telephone System Model:

1. "TELEPHONE SYSTEM MODEL-1"
2. PREAMBLE
3. NORMALLY, MODE IS INTEGER
4. EVENT NOTICES INCLUDE ARRIVAL AND CLOSING
5. EVERY DISCONNECT HAS A DIS-CALL
6. TEMPORARY ENTITIES
7. EVERY CALL HAS AN ORIGIN AND A DESTINATION
8. DEFINE LINK-IN-USE, MAX-LINKS, BLOCKED, BUSY, FINISHED
9. AND STOP-TIME AS VARIABLES
10. DEFINE INTER-ARRIVAL-TIME AND MEAN-LENGTH AS REAL VARIABLES
11. PERMANENT ENTITIES
12. EVERY TLINK HAS A STATE
13. DEFINE SECS TO MEAN / 60 MINUTES
14. END

CHURCHILL

Pg 256
THE MAIN ROUTING
(Defining First Figure)
of Telephone System

Date: 30th May Page: _____

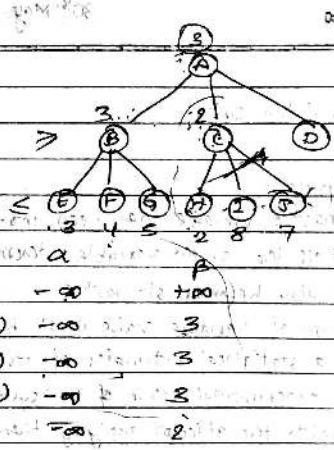
Analysis of Simulation Output

① Nature of the problem:

- Once the stochastic variable has been introduced into a simulation model, all the system variable describing the system behaviour also becomes stochastic.
- Many observation of variable value must be made in order to make a statistical estimates of its true value
- Because of the experimental nature of the system simulation, it seems natural to the attempt applying these methods to simulation results.

(From Photocopy Paper Remaining)

CHURCHILL



Simulation Run Statistics:-

Let us see an examples that explains this analysis in simple way of simple single server system.

- single server → service time → exponential dist.
- FIFO Queuing discipline → inter-arrival time → exponential

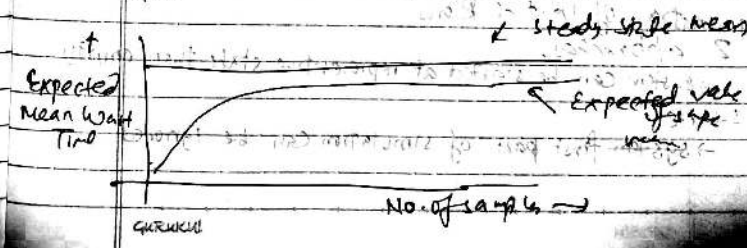
In simulation run the simplest way to find sample mean.

$$\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i \quad (x_i, i=1, 2, \dots, n)$$

$$\text{as } \bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i \quad i=1, 2, \dots, n$$

2 problems:-

- ① → waiting time is not independent coz when waiting time line forms, waiting time depends on the waiting time of predecessor
- ② Distribution may not be stationary, coz simulation run is started with the system in some initial state, then in idle state in which no customer waiting & thus early arrivals have high probability of obtaining service quickly. Sample mean therefore includes the early arrivals which is biased.



Replication of runs:-

→ One problem during simulation run is that while we are measuring statistics the results are dependent.

→ But in simulation indep result is required.

→ So to obtain indep result is by repeating the simulation.

Simulation

→ Repeating same experiment with diff random no of size n give independent result of sample $\bar{x}_j(x)$.

→ Suppose:

x_{ij} = i th observation for j th run

Then $\bar{x}_j(n) = \frac{1}{n} \sum_{i=1}^n x_{ij}$

Then variance is

$$s_j^2(n) = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j(n))^2$$

Now combining results of p indep runs

give

$$\bar{x} = \frac{1}{p} \sum_{j=1}^p \bar{x}_j(n) \quad \bar{x}^2 = \frac{1}{p} \sum_{j=1}^p \bar{x}_j^2(n)$$

$$s^2 = \frac{1}{p} \sum_{j=1}^p s_j^2(n) \quad s^2 = \frac{1}{p} \sum_{j=1}^p s_j^2(n)$$

Elimination of Initial Run

2 approaches.

→ system can be started at representative state than empty state.

→ system first part of simulation can be ignored.

① It is necessary to know the steady state distribution. so we can select initial state distribution. In simulation process it is necessary to select better initial condition, coz there may be information available on expected condn. Thus removing initial bias.

→ 2nd part is common approach.

→ Here, the first part of initial section is removed containing highly biased result is eliminated. The process starts at an idle state and continues until a certain period of time. The entities existing upto that pt are left as they are and this pt is the starting pt of the system to repeat simulation run & statistic is also kept gathering from here.

Disadv:

→ No simple rules can determine how long to an interval should be eliminated.

Simulink Lib.

Sink $\rightarrow$ O/P, Source $\rightarrow$ S/P.

File $\rightarrow$ New $\rightarrow$ Model

Source $\rightarrow$ clk.

Commonly used blocks $\rightarrow$ gain.

gain $\rightarrow$ 2 t. (2) Parameters.

math operation $\rightarrow$ trigonometric function

CO

Commonly used blocks $\rightarrow$ sum.

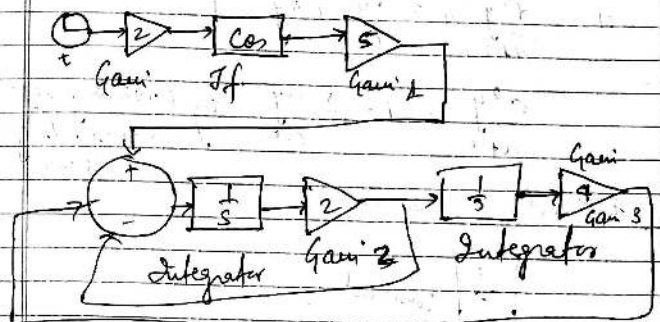
Commonly used blocks $\rightarrow$ $\frac{1}{s}$ Integrator

Sink $\rightarrow$ scope $\rightarrow$ O/P.

Run $\rightarrow$ Scope double click.

To change the graph click in graph and Autoscale.

$$\frac{d^2x}{dt^2} = 5 \cos(2t) - 2 \frac{dx}{dt} = 4d, \text{ p.s.k.m.}$$



Random Nois Congruential method.
3 type.

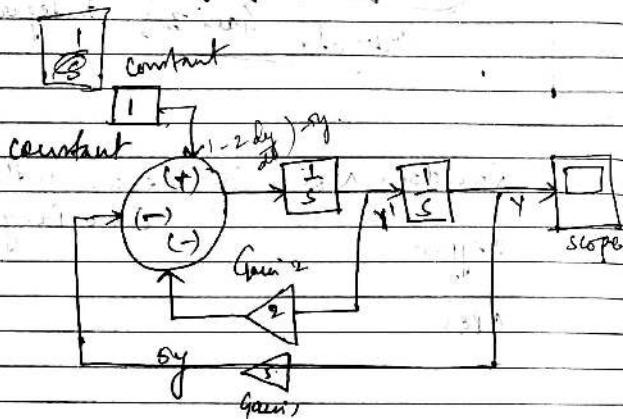
GPS

Integration - $\frac{1}{s}$

$$1) \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} + 5y = 1$$

$$\frac{d^2y}{dt^2} = 1 - 2 \frac{dy}{dt} - 5y \quad (\text{rearrange eqn for highest derivative})$$

2nd order $\rightarrow$ 2 Integrators



eliminate END & START

file menu, program

1) Edit $\rightarrow$ Insert ^{block} $\rightarrow$ To display generated program

2) Command $\rightarrow$ Create Simulation Ready (By default should be shown)

3) windows $\rightarrow$ simulation windows $\rightarrow$ Block window

Table
Command Start 100 (value) (no. of times it will run)

$\downarrow$
app screen analysis

4 Apr 1

4 Apr 1

2) Mixed Congruential method of generating random numbers a, b & m are constants, the values of which are suitably selected and entered. n is the random numbers to be generated. Seed is the starting random no. which is also to be entered.

```

→ int a, b, k, i, j, m, n, seed, r[50];
pf ("Enter int values for a, b, m = ");
sf ("%d %d %d", &a, &b, &m);
pf ("Enter SEED value = ");
sf ("%d", &seed);
pf ("Enter no. of random no. to be generated = ");
sf ("%d", &n);
r[0] = seed; // a = 2, b = 53, m = 1000
for (i = 1; i <= n; i++)
{
    r[i] = (a * r[i-1] + b) % m;
    printf ("%4d", r[i]);
}
getch ();

```

012

145	98	111	384	117	570	768	76649
682	375	918	541	414	798	740	599506
679	312	605					
936	709	712					
277	366	789					
24000	83	716					

Generate 4,0
 Queue system
 Arrive in queue
 Serve inspector
 Depart in queue
 Advance 5,2
 Release Inspector
 Depart system
 Transiger 0.15, Acc, Rij
 Acc Generate 1
 Rij Terminate 1

Chapter-4

ANALYSIS OF SIMULATION OUTPUT

In the previous chapters we have learn how to produce a model of a system, how to simulate and produce a computer program for simulating the system such as predator prey system. By which of this program, we can performed the desired experiment on the system being studied and can analyze the output. In those programs, we have a freedom to specify initial condition, run length or replication runs in order to analyze the output simulation.

A large body statistical method has been developing over the years to analyze results in science, engineering and other fields where experimental observation are made. So, because of experimental measure of system of simulation for these statistical methods can be adapted to simulation results to analyze.

The newly developing statistical methodology concerns;

- 1) To ensure that the statistical estimates are consistent, meaning that as the sample size increases the estimate tends to true value.
- 2) To control biasing in measure of both new values of variance. Bias causes the distinction of estimate to differ significantly from the true population statistics, even though the estimate may be consistent.
- 3) To develop sequential testing methods, to determine how long a simulation should be run in order to obtain a confidence in its return.

ESTIMATION METHOD

Statistical methods are commonly used on random variable. Usually a random variable is drawn from an infinite population with a finite mean ' μ ' and finite variance ' σ^2 '.

These random variable are independently and identically distributed (i.e. iid variables).

Let, x_i = iid random variables, $i=1, 2, \dots, n$, then according central limit theorem and applying transformation, approximate normal variance,

$$Z = \frac{\sum_{i=1}^n x_i - n\mu}{\sqrt{n\sigma^2}}$$

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Var. In Prob. theory & statistics, Var. is the expectation of the squared deviation of a random variable from its mean and it informally measures how far a set of random variables are spread out from their mean.

Where $\bar{x}$ = sample mean

It can be shown to be a consistent estimator for the mean of population from which the sample is done.

Since the sample mean is some of random variable, it is itself a random variable. So a confidence interval about its computed value needs to be established.

The probability density function on the standard normal variable (Z) is shown in figure (1).

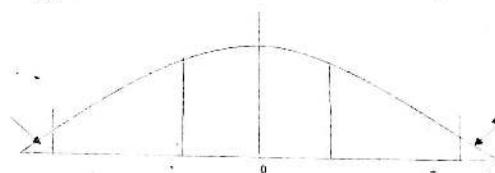


Fig. (1)

The integral formed $-\infty$ to p is the probability that Z is less than or equal to p and is denoted by $\Phi(u)$.

Let us consider the value of u (u_α) such that $\Phi(u) = 1 - \alpha/2$ where α = some constant < 1 .

Then probability of Z for $Z > u_\alpha$ = $\alpha/2$.

Probability of Z for $-u_\alpha < Z < u_\alpha$ = $1 - \alpha$.

In terms of sample mean $\bar{x}$, the probability statement can be written as

$$\text{Prob} \left\{ \bar{x} + \frac{\sigma}{\sqrt{n}} u_\alpha \geq \mu \geq \bar{x} - \frac{\sigma}{\sqrt{n}} u_\alpha \right\} = 1 - \alpha$$

Here the constant $1 - \alpha$ is a confidence level (usually expressed in %) and the interval

$$\bar{x} \pm \frac{\sigma}{\sqrt{n}} u_\alpha$$

is the confidence interval.

Estimation population variance s^2 (not σ^2 which is actual population variance) is given by:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Replacing $\frac{u_a^2}{2}$ by $\frac{t_{n-2, \frac{\alpha}{2}}^2}{2}$, then the estimated variance s^2 , the confidence interval for $\bar{x}$ is given by

$$C = \bar{x} \pm \frac{s}{\sqrt{n}} t_{n-2, \frac{\alpha}{2}}$$

SIMULATION RUN STATISTICS

On every simulation run, some statistic are measure based on some assumption, for example - on establishing confidence interval it is assume that the observation are mutually independent and distinction from which they are drawn is stationary. But many statistics are interest in a simulation don't meet this condition.

Let us illustrate the problems that arise in measuring statistic from simulation run with the example of single server system.

Consider occurrence of arrivals has a Poisson distribution,

- The service time has an exponential distribution.
- The queuing discipline is FIFO
- The inter-arrival time is distributed exponentially
- System has a single server.

Then in a simulation run, the simplest way to estimate the mean waiting is to accumulate the waiting time of n successive entities and dividing it by ' n '. This gives sample mean denoted by $\bar{x}(n)$.

If x_i ($i = 1, 2, 3, \dots, n$) are the individual waiting times, then

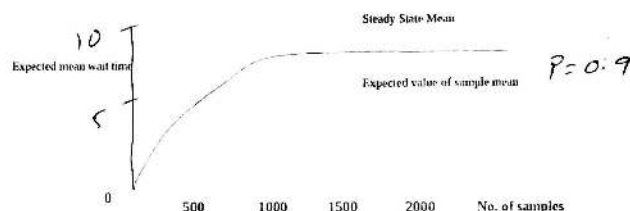
$$\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$$

The 1st problem, here is that, the waiting times measure this way are not independent because whenever a waiting line forms, the waiting time of each entity on the line depends upon the waiting time of its predecessor (i.e. the entities are auto co-related).

The usual formula for estimating the mean value of the distribution remains on satisfactory estimate for the mean of auto co-related data. However the variance of auto-correlated data is not

related to the population variance by simple expression $\frac{\sigma^2}{n}$ as occurs for independent data.

The 2nd problem is that the distribution may not be stationary; it is because a simulation run is started with the system in some initial state, frequently the idle state, in which no service is being given and no entities are waiting, thus the early arrivals have a more probability of obtaining service quickly. So a sample mean that includes the early arrivals will be biased. As the length of simulation run extended and the sample size increases, the effect of bias will be minimum. This is shown in below fig.



REPLICATION OF RUNS

One problem in measuring the statistic in the simulation run is that the results are dependent. But it is required, in simulation, to get the independent result. The one way of obtaining independent result is to repeat the simulation.

Repeating the experiment with different random numbers for the same sample size ' n ' gives a set of independent determination of sample mean $\bar{x}(n)$.

Even though the distribution of sample mean depends upon the degree of auto correlation, this independent determination of sample mean can be used estimate the variance of distribution. Suppose,

- Experiment is repeated p-times with independent random numbers,

- $x_0 = i^{th}$ observation of j^{th} run. Then estimates for:

- Sample mean $\hat{x}_j(n) = \frac{1}{n} \sum_{i=1}^n x_{ji}$

- Then estimate for variance for j^{th} run $s_j^2(n)$ is given by

$$s_j^2(n) = \frac{1}{n-1} \sum_{i=1}^n (x_{ji} - \hat{x}_j(n))^2$$

Now, combining the results of p independent measurement gives the following estimate for the mean waiting time $\hat{x}$, and variance s^2 :

$$\hat{x} = \frac{1}{p} \sum_{j=1}^p \hat{x}_j(n)$$

$$s^2 = \frac{1}{p} \sum_{j=1}^p s_j^2(n)$$

Here, the value of $\hat{x}$ is an estimate for mean waiting time and the value of s^2 can be used to establish a confidence of intervals.

ELIMINATION OF INITIAL BIAS

There two general approaches that can be used to remove the initial bias;

- 1) The system can be started in more representative states rather than in the empty state.
- 2) The first part of the simulation run can be ignored.

- In the first approach, it is necessary to know the steady state distinction for the system and we then select the initial state distinction. In the study of simulation, particularly existing system, there may be information available on the expected condition which makes it feasible to select better initial condition and thus eliminating the initial bias.

- The second approach that is used to remove the initial bias is the most common approach. In this method, the initial section of the run which has highly bias (simulation) result is eliminated. First, the run is started from an idle state and stopped after a certain period of time (the time at which the bias is satisfactory). The entities existing in the system at that are left as they are and this point is the point of restart for other repeating simulation run. Then the run is restarted with statistics being gathered from the point of restarted. These approaches have following difficulties:

- 1) No simple rules can be given to decide how long an interval should be eliminated. For this we have to use some pilot run starting from the ideal state to judge how long the initial bias remains. These can be done by plotting the measured, statistics against the run length.
- 2) Another disadvantage of eliminating the first part of simulation run is that the estimate of variance will be based on less information affecting the establishing of confidence limit. These will then cause to increase in confidence interval size.